

### Evoke QA4

Motorola's new messaging phone, the Evoke QA4, is the latest high profile handset to incorporate OKLabs virtualisation technology

### Xen

The open source virtualisation project, has incorporated support for ARM mobile processors

the virtual machine may be a smartphone operating system for complex tasks and applications running in parallel with a real-time operating system (RTOS) which takes care of the regular phone functions. There are different types of virtualisation being implemented and tested by different vendors.

The most basic form of virtualisation is actually application virtualisation which is almost the same concept as the Java runtime environment. Here, the hypervisor software runs on top of a normal operating system and supports different applications running on top of the hypervisor. The advantage of this approach is that the developer does not need to develop separate applications for each platform or operating system since the hypervisor takes care of all interactions with the base operating system and hardware. Another advantage is that this can run on basic feature phones and provide advanced applications to users which are normally provided by smartphones.

Cloud-based virtualisation is also gaining ground these days. With the advent of cloud computing and more and more enterprises moving data and applications to the cloud, mobile virtualisation can also take advantage of this technology. Cloud-based virtualisation involves a virtual client interface running on the mobile phone where all inputs are given while all the actual processing and storage actually occurs in the cloud. Cloud computing minimises the need for powerful mobile processors since only display capability is required and also reduces power consumption. A major advantage of cloud computing is data security since the actual application and all the data resides on the cloud which can be secured in a much better manner than cell phones storing the same data. However, cloud based virtualisation requires a constant online connection which may not be possible at all places.

Virtualisation is also used for gaining remote access to machines running other operating systems such as Windows. This may be considered an extension of the cloud based virtualisation model since we use the application to display data and information being processed at some other place.

The most useful kind of virtualisation is the one most commonly used in servers around the world. Multi-OS virtualisation is used to execute multiple operating systems on a hypervisor running on a common piece of hardware. The operating systems can either be running side by side or the machine may need a reboot to switch between the two operating systems depending on the implementation. In the rest of the article, we are going to look in more detail at multi-OS virtualisation in mobile environments.

Mobile virtualisation is useful for everyone, starting from handset manufacturers to application developers and end users.

A major application of virtualisation is in terms of security. Users can have one secure virtual machine for business applications and a second one for personal use with ringtones, song, games etc. The advantage of this setup is that the secure virtual machine is protected from any issues occurring due to the personal virtual machine. Any malware or crash will only affect the personal virtual machine leaving the secure one intact.

With the help of mobile virtualisation, users can also have multiple profiles on their phones, one for business use and one for personal use. This is particularly helpful for users who usually carry multiple phones with them to office etc. Depending on the particular implementation, the user may be able to use



**"Your phone is becoming a combination of your PC and wallet all rolled into one. This could become the most important device a consumer can own."**

Srinivas Krishnamurti  
Director of Product Management and  
Market Development, VMware Inc

both profiles together or login into one profile at a time. This will also allow the user to bring a device of his choice to work since he can then use a secure corporate connection for business related activities.

This will also be helpful to corporate IT departments in their standardisation processes since they can load the same secure operating system image on all phones without any hardware differences. Also, in case of any issues with the virtual machine, IT can simply wipe the secure virtual machine and copy a new one in its place without affecting any personal data stored on the device in other virtual machines.

Since its now possible to do more and more activities using smartphones, an increasing number of people are keeping their personal

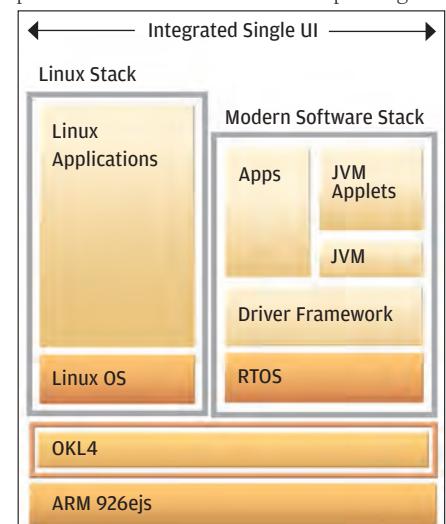
data on their phones. In case of mobile virtualisation, all personal data is present in the virtual machine. So, any user upgrading to a new smartphone can simply copy his virtual machine to the new device, thus avoiding any issues with regards to applications or data.

The manufacturer can offer phones with a choice of operating systems since the operating system is just a virtual machine which can be installed at the point of sale. This increased choice to the user may indirectly increase competition between operating systems while giving more options to the user.

Mobile virtualisation also results in cost savings for manufacturers in terms of the hardware used in the device. Instead of separate microprocessor chips for each function, virtualisation and the software hypervisor enable a single core chip to perform all the different functions which are split into separate virtual machines.

Mobile virtualisation also translates into a shorter time to market for operating system developers. This is because for traditional smartphones, every new phone has a different chipset and so the operating system for each new model must be tweaked for the specific hardware in the device. With mobile virtualisation, the same operating system can be used on multiple devices since the hypervisor takes care of any differences in the hardware for each new model.

Application development becomes a much simpler process for developers in case mobile virtualisation is being used. This is because they will only need to develop their application for one platform. Using mobile virtualisation, the application can automatically be used on other smartphone platforms as well. e.g. an application written for the Android operating system can be run on a Windows Mobile smartphone using mobile virtualisation which will run the Android operating system in parallel with the windows mobile operating



Mobile Virtualisation: Motorola Evoke Hardware Structure